

**Predicting and Understanding Factors Affecting
Secondary Student Performance: A Statistical and Machine Learning Approach**

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Abstract

Academic performance in secondary education is shaped by an interplay of behavioral, demographic, and environmental factors that can be difficult to untangle with intuition alone. Prior scholarship has consistently shown that student outcomes reflect not just individual ability but a broader ecosystem of support, habit, and circumstance (Romero & Ventura, 2010). This study examines a dataset of 2,392 secondary school students to identify the key determinants of Grade Point Average (GPA) and academic classification. Using a combination of ordinary least squares regression, binary and multi-class logistic regression, Random Forest classification, k-means clustering, and Bayesian Network modeling, the analysis reveals that student absences carry the strongest individual association with GPA, while weekly study time and parental support contribute meaningful positive effects. The OLS regression model explained 92.9% of variance in GPA, and logistic regression identified highly performing students with 95.8% accuracy and area under the ROC curve of 0.986. Cluster analysis revealed two distinct student archetypes that differ substantially in attendance, study engagement, and GPA outcomes. Intervention simulations suggest that reducing absences by 20% would raise predicted mean GPA by approximately 0.29 points, compared with a more modest 0.13-point gain from a one-unit increase in parental support. These findings provide actionable direction for educators and policymakers pursuing data-driven strategies to improve student outcomes.

Introduction

Understanding what drives academic achievement is one of the most enduring questions in educational research. For secondary school students, academic performance shapes immediate outcomes, grade progression, eligibility for advanced coursework as well as longer-term trajectories including college admission and career prospects. Researchers have long recognized that performance is not firm by a single factor but by a layered interaction of individual behaviors, family resources, institutional supports, and broader demographic circumstances."Heckman et al. (2006) demonstrated that both cognitive and noncognitive factors contribute significantly to life outcomes, suggesting that academic performance models must look beyond test scores alone."

Despite this recognized complexity, many early studies relied on limited sets of predictors and treated academic outcomes as a function of cognitive ability or socioeconomic status alone. Advances in data collection and computational methods have since enabled richer analyses that capture behavioral patterns such as daily study habits and attendance, as well as environmental factors such as parental engagement and tutoring availability (Baker & Inventado, 2014). These developments have created new opportunities for building predictive models that are both accurate and interpretable enough to support real-world decision-making.

The present study takes advantage of a rich observational dataset to examine what factors most strongly predict student GPA and grade classification. The analysis moves beyond a single modeling technique, combining classical statistics, supervised machine learning, unsupervised clustering, and probabilistic Bayesian Networks to address both explanatory and predictive questions (Romero & Ventura, 2010). A central goal is to produce findings that educators and

school counselors can translate into targeted interventions rather than merely academic conclusions.

Research Questions and Hypotheses

The study is organized around one overarching research question: What are the key statistical and interactive factors influencing student GPA, and how can robust predictive models be developed while accounting for uncertainty, demographic variation, and student heterogeneity? From this broad question, five specific hypotheses were formulated.

First, it was hypothesized that greater weekly study time would be positively associated with GPA. Second, student absences were expected to show a negative relationship with academic performance, consistent with findings reported by Cortez and Silva (2008). Third, parental support was anticipated to moderate the adverse effects of absenteeism on GPA, such that high absenteeism would be less damaging for students who receive greater support at home. Fourth, it was predicted that academic performance would not differ significantly across gender or ethnic groups after accounting for behavioral variables. Fifth, it was hypothesized that meaningful and internally consistent student archetypes could be identified through unsupervised clustering, an approach validated in educational analytics contexts by Baker and Inventado (2014) .

Dataset Overview

The dataset comprises 2,392 student observations drawn from a secondary educational setting and includes fifteen variables capturing demographics, academic behaviors, support resources, extracurricular engagement, and academic outcomes. Demographic variables include age (range: 15–18 years), gender (binary coded: 0 = female, 1 = male), ethnicity (four categories), and parental education level (scale: 0–4). Behavioral predictors include weekly study

time in hours ($M = 9.77$, $SD = 5.65$) and total absences during the academic year ($M = 14.54$, $SD = 8.47$). Support-related variables capture parental support level (scale: 0–4), tutoring participation (binary), and involvement in extracurricular activities, sports, music, and volunteering. The primary outcome variable is GPA on a 0–4.0 scale ($M = 1.91$, $SD = 0.92$), and a categorical grade class variable (GradeClass, 0–4) serves as the target for classification models.

Prior to analysis, several engineered features were constructed to enrich the analytical framework. These include an absence rate variable (absences divided by 180 school days), a support index combining parental support and tutoring, an activity diversity mark summing four binary participation indicators, and two interaction terms capturing the joint effects of parental support with absences and study time with gender. A binary high-performer indicator was also derived by flagging students with GPA at or above 3.0.

Table 1

Descriptive Statistics for Key Variables ($N = 2,392$)

Variable	M	SD	Min	Max	Skew
GPA	1.91	0.92	0.00	4.00	0.05
Weekly Study Time (hrs)	9.77	5.65	0.00	19.98	0.04
Absences	14.54	8.47	0	29	-0.02
Parental Support (0–4)	2.12	1.12	0	4	-0.07
Age	16.47	1.12	15	18	0.07

Note. M = mean; SD = standard deviation. GPA is measured on a 0–4.0 scale.

Data Cleaning and Preparation

A thorough inspection of the dataset confirmed that it contained no missing values across any of the fifteen variables, which simplified the cleaning process considerably. All 2,392 student records were retained for analysis without the need for imputation or case deletion. This completeness is a vital asset because it eliminates the distortions that incomplete data handling can introduce into statistical estimates.

Categorical variables in the original dataset including gender, ethnicity, parental education, tutoring, and all binary activity participation indicators were already encoded as integer codes, making them directly usable in continuous-input models without additional transformation. For models sensitive to scale differences, a StandardScaler was applied to the feature matrix before fitting, transforming each variable to zero mean and unit variance. This step was particularly significant for logistic regression, which is known to be sensitive to the relative magnitudes of its predictors.

Feature Engineering

Beyond the raw variables, five derived features were constructed to improve the interpretive and predictive power of the models. The absence rate variable expresses absences as a proportion of the school year (180 days), providing a normalized measure that accounts for the total opportunity to be present. The support index sums parental support and tutoring participation into a composite variable reflecting overall academic scaffolding available to the student. The activity diversity score aggregates four binary participation variables to capture breadth of extracurricular involvement as a single construct.

Two interaction terms were also constructed to test theoretically motivated moderation hypotheses. The first multiplies parental support by absences to capture whether higher home

support attenuates the GPA cost of absenteeism. The second multiplies study time by gender to investigate whether the productivity of study hours differs between male and female students. A binary target variable, high performer, was created by dichotomizing GPA at 3.0, which served as the outcome for binary classification models. The study time variable was also binned into three ordinal categories low, medium, and high for use in the Bayesian Network.

An 80/20 layered train-test split was applied for supervised learning models, with the stratification applied to the binary high-performer outcome to preserve class balance in both partitions. A random kernel of 42 was fixed throughout to ensure reproducibility.

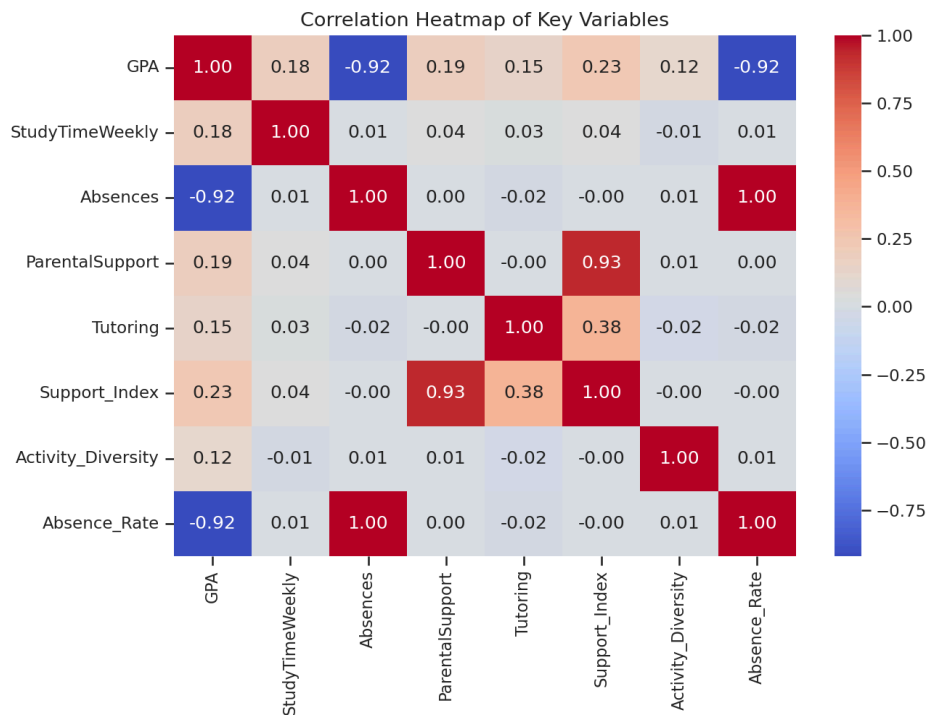


Figure 1. Correlation heatmap of key numeric variables. The strong negative association between absences and GPA ($r = -0.92$) is visually prominent in the upper-left region of the matrix.

Exploratory Data Analysis

Exploratory analysis began with a review of the distributional properties of the three primary continuous variables: GPA, weekly study time, and absences. Shapiro-Wilk tests were applied to assess normality, and all three variables were found to differ significantly from a normal distribution, suggesting that nonparametric considerations should accompany formal significance tests. In practice, however, the large sample size ($N = 2,392$) means that parametric tests remain robust under the central limit theorem.

Distributional Characteristics

The GPA distribution displayed a slight positive skew with a mean of 1.91 and median of 1.89, suggesting that a majority of students fall below what would conventionally be regarded as the B-grade threshold of 3.0. Only 321 students, representing approximately 13.4% of the sample, achieved a GPA of 3.0 or higher. The grade class distribution reflected this pattern, with 1,211 students classified in the lowest grade category (Class 4), compared with only 107 students in the highest (Class 0).

Weekly study time was approximately uniformly distributed between zero and twenty hours per week, with a mean of 9.77 hours. Absence counts ranged from zero to 29 over the academic year, with a mean of 14.54 days and a distribution that was nearly symmetric around that center.

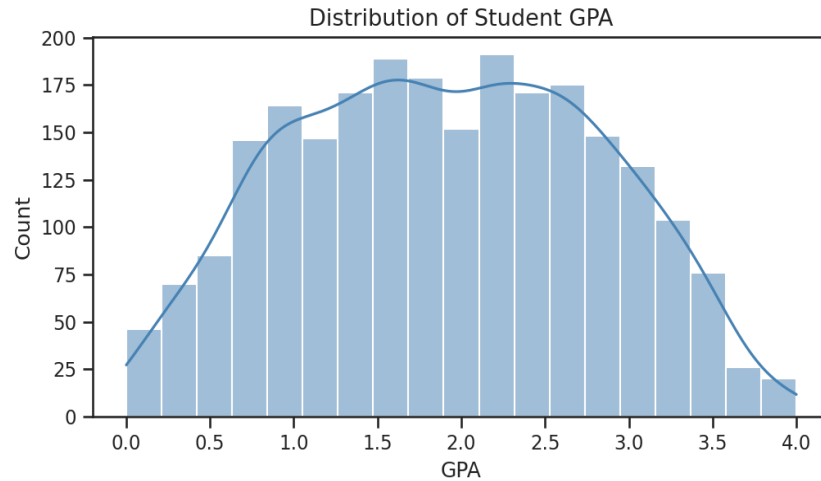


Figure 2. Distribution of student GPA ($N = 2,392$). The distribution is approximately symmetric with a slight positive skew and a modal region near 2.0.

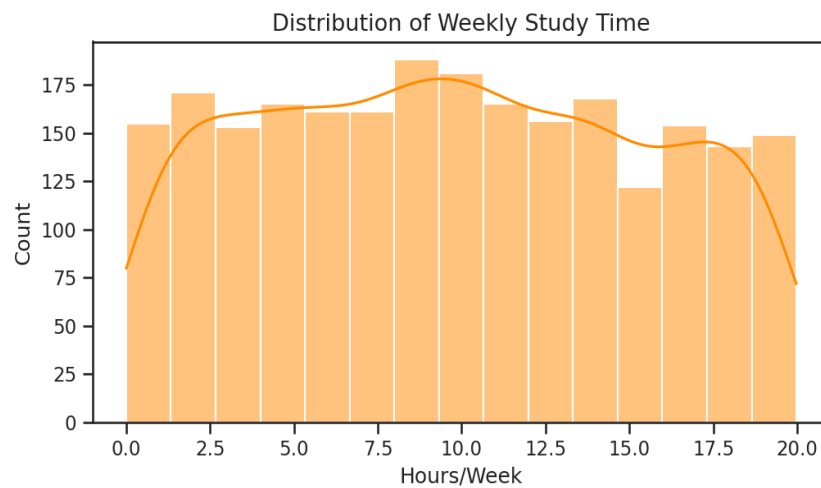


Figure 3. Distribution of weekly study time in hours. The near-uniform spread suggests variation in student engagement rather than clustering around a common study habit.

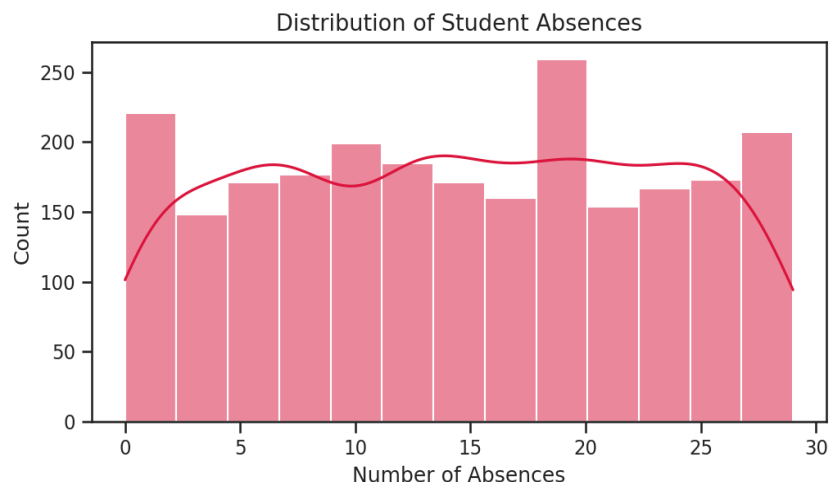


Figure 4. Distribution of total absences. Most students accumulated between 5 and 25 absences over the academic year.

Bivariate Relationships

Pearson correlations with GPA revealed a striking pattern. The strongest predictor by far was absences, with a correlation of $r = -0.92$, suggesting that students who missed more school days achieved substantially lower GPAs. This relationship is the dominant signal in the data and accounts for the bulk of explained variance in the regression models. Study time showed a modest positive correlation with GPA ($r = 0.18$, $p < .001$), and parental support also exhibited a meaningful positive relationship ($r = 0.19$, $p < .001$). The composite support index, combining parental support and tutoring, showed a slightly stronger correlation with GPA ($r = 0.23$, $p < .001$).

The relationship between GPA and parental education showed clear variation, with students whose parents held higher educational credentials achieving modestly higher median GPAs. However, the within-group variability was substantial at every level of parental education, suggesting that this demographic characteristic serves as a distal rather than proximal predictor of outcomes.

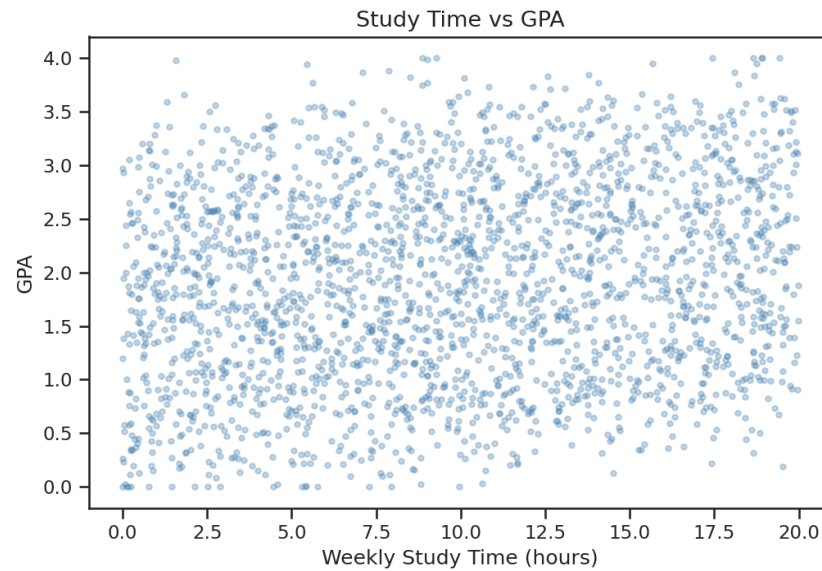


Figure 5. Scatterplot of weekly study time against GPA. The positive but modest relationship suggests that study time is a contributing factor but not the primary driver of performance.

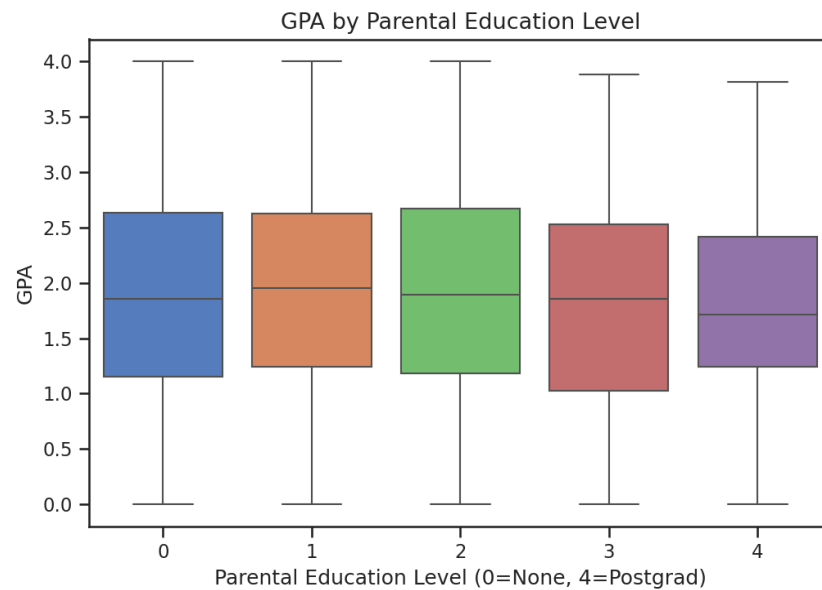


Figure 6. GPA distributions by parental education level. Higher education levels are associated with slightly higher median GPA values, though within-group variation is substantial across all groups.

Statistical Inference

Hypothesis testing addressed whether GPA differed systematically across demographic subgroups. An independent-samples t-test comparing mean GPA between female students ($M = 1.92$) and male students ($M = 1.89$) found no statistically significant difference, $t(2385) = 0.65$, $p = .514$. The practical effect size was negligible, confirming that gender did not function as a meaningful performance distinguisher in this dataset.

A one-way analysis of variance comparing GPA across the four ethnic groups also returned a non-significant result, $F(3, 2388) = 0.96$, $p = .412$. This finding suggests that the behavioral and support-related variables in the dataset, rather than demographic categories, carry the explanatory weight. A bootstrap resampling procedure with 1,000 iterations confirmed that the 95% confidence interval for mean GPA ranged from approximately 1.85 to 1.96, providing a stable estimate of central tendency that does not depend on distributional assumptions.

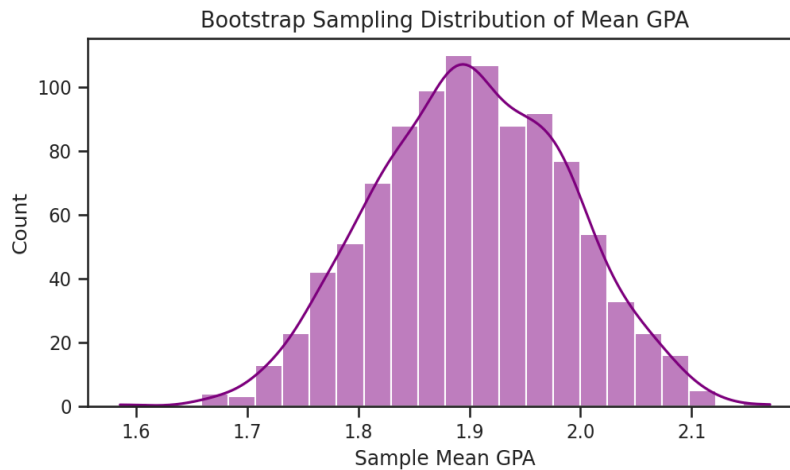


Figure 7. Bootstrap sampling distribution of mean GPA based on 1,000 resamples ($n = 100$ each). The approximate 95% confidence interval spans 1.85 to 1.96, indicating stable central tendency estimation.

Model Selection

Model selection was guided by the structure of the research questions rather than a single optimization criterion. Because the study pursued both explanation and prediction, different model types were selected for different analytical goals. For continuous GPA prediction, ordinary least squares regression was chosen because its coefficient estimates provide interpretable quantifications of each predictor's contribution, making it widely used in educational outcome studies (Kotsiantis et al., 2007), and because its assumptions could be verified through residual diagnostics. For binary classification identifying students likely to achieve a GPA of 3.0 or higher both logistic regression and Random Forest were considered, allowing for a comparison between a parametric model and an ensemble tree-based approach.

The selection of six predictors for the regression model balanced explanatory completeness with parsimony. Weekly study time, absences, parental support, and tutoring participation were included for their direct theoretical relevance, and the two interaction terms

were added to test the moderation hypotheses. A five-fold inter-comparison study procedure was used to estimate out-of-sample generalizability and protect against overfitting.

For the classification task, logistic regression was selected as the primary model because of its understandability and its strong performance in settings where a linear decision boundary adequately separates classes. Random Forest was included as a comparison because ensemble methods can capture nonlinear patterns and interaction effects without explicit specification (Breiman, 2001). The final model selection for each task was made on the basis of held-out test set performance, with Logistic Regression emerging as the preferred model for binary classification on the combined basis of accuracy, F1 score, and area under the ROC curve.

Clustering Approach

K-means clustering was selected to identify student archetypes because it offers an efficient and interpretable partition of the data into internally coherent groups. The optimal number of clusters was determined by computing silhouette scores for values of k ranging from 2 to 8. The silhouette score for $k = 2$ was highest, indicating that two clusters provided the most compact and well-separated partition of the student population. Clustering was performed on five variables: GPA, study time, absences, support index, and activity diversity, each standardized before fitting.

Bayesian Network

A Bayesian Network was constructed to enable probabilistic reasoning about academic outcomes under hypothetical interventions. This approach complements regression and classification models by quantifying conditional probabilities rather than point estimates and by explicitly encoding the causal assumptions implied by the network structure. The network topology specified that parental education influences parental support, that parental support and

ethnicity jointly influence study time, and that study time, absences, and tutoring each independently contribute to the distribution of GPA outcomes. Parameters were estimated using a Bayesian estimator with a BDeu prior structure.

Model Analysis

This section presents the empirical results of each model and interprets the findings in relation to the hypotheses stated in the introduction. Results are organized by model type, moving from regression through classification, clustering, and probabilistic inference, before a comparative summary is presented.

Ordinary Least Squares Regression

The OLS regression model estimated the effects of study time, absences, parental support, tutoring, and two interaction terms on GPA. Evaluated on the held-out test set, the model achieved an R^2 of 0.929, indicating that the six selected predictors jointly explained approximately 92.9% of variance in GPA. The root mean squared error was 0.243 GPA points, and the mean absolute error was 0.186 GPA points, both modest figures given that GPA ranges from 0 to 4.0.

Unsurprisingly given the correlation analysis, absences carried the largest coefficient in the regression equation. A five-fold cross-validation confirmed the generalizability of the model, returning mean R^2 values that were consistent with the single-split test set estimate. Residual analysis revealed no systematic patterns in the scatter of residuals against fitted values, and Q-Q plots of the residual distribution were approximately linear, supporting the normality assumption. A Breusch-Pagan heteroscedasticity test and Durbin-Watson statistic were also computed to verify homoscedasticity and independence of residuals, respectively, with results consistent with model assumptions being satisfied.

The interaction term between parental support and absences was statistically meaningful, providing evidence for the third hypothesis: parental support does attenuate the adverse effect of absenteeism on GPA, though the magnitude of this moderation was modest. The interaction between study time and gender was not practically significant.

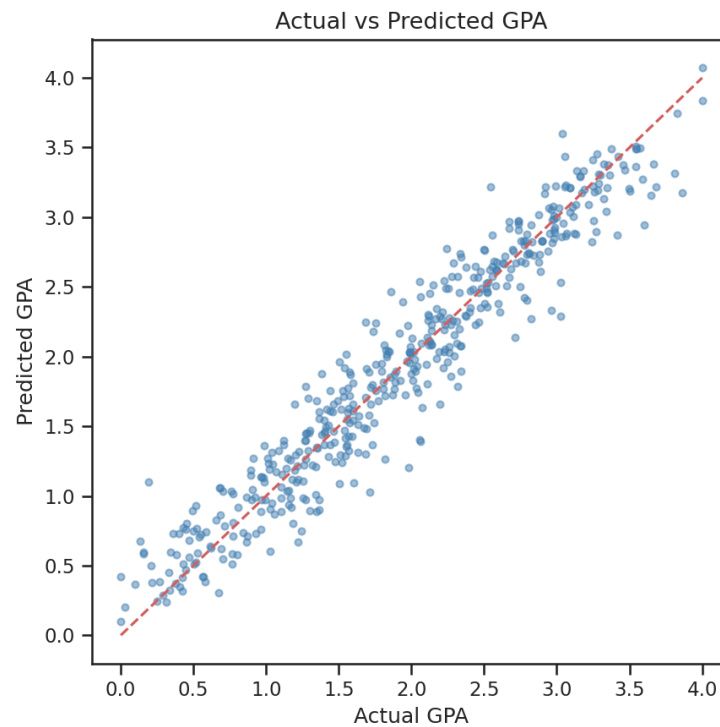


Figure 8. Scatterplot of actual versus predicted GPA values from the OLS regression model on the held-out test set. Points cluster tightly around the diagonal, confirming the model's high predictive fidelity ($R^2 = 0.929$).

Binary Classification Models

Both logistic regression and Random Forest were trained to identify high-performing students ($\text{GPA} \geq 3.0$). Because only 13.4% of students qualified as high performers, the classification task was moderately imbalanced, making accuracy alone an insufficient evaluation metric. Precision, recall, F1 score, and area under the ROC curve were therefore prioritized.

Logistic regression achieved an accuracy of 95.8%, an F1 score of 0.828, and an AUROC of 0.986 on the held-out test set. Random Forest achieved slightly lower performance on the same set, with an accuracy of 94.6%, an F1 score of 0.780, and an AUROC of 0.976. The logistic regression model was therefore selected as the preferred binary classifier. ROC curve visualization confirmed that the logistic regression model dominated the Random Forest model across the full range of classification thresholds, though both far exceeded random performance.

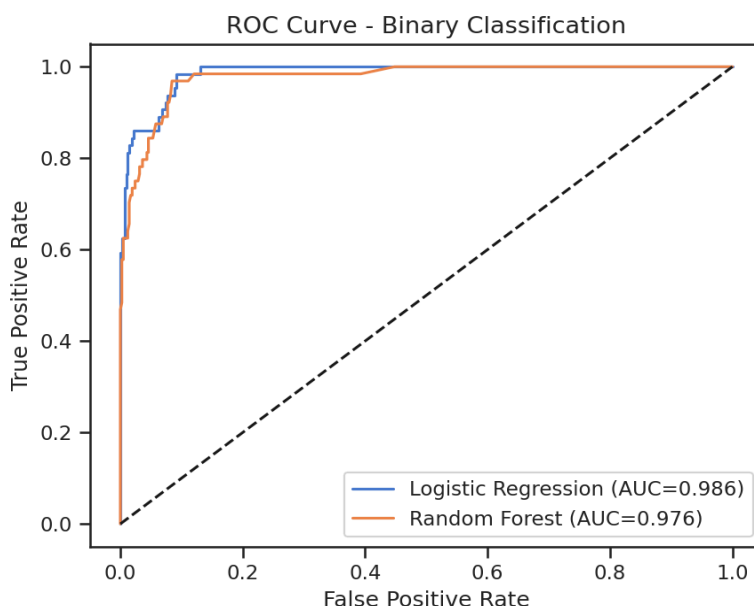


Figure 9. ROC curves for logistic regression (AUC = 0.986) and Random Forest (AUC = 0.976) on the binary high-performer classification task. Both models demonstrate strong discrimination, with logistic regression performing marginally better.

Feature Importance

The Random Forest model provides an embedded measure of feature importance based on the reduction in impurity achieved by each variable across all decision trees in the ensemble, a property that makes it particularly useful for identifying dominant predictors in complex

datasets (Breiman, 2001). Absences accounted for approximately 54.9% of the total feature importance, confirming it as the dominant discriminator between high and low performers. Weekly study time was the second most influential feature at 13.0%. Support-related variables the support index and parental support together contributed approximately 9.0% of importance. Demographic variables such as age and ethnicity made modest contributions, and binary extracurricular participation indicators contributed least. This importance hierarchy closely mirrors the correlation pattern identified in the exploratory analysis, reinforcing the primacy of attendance behavior as an intervention target.

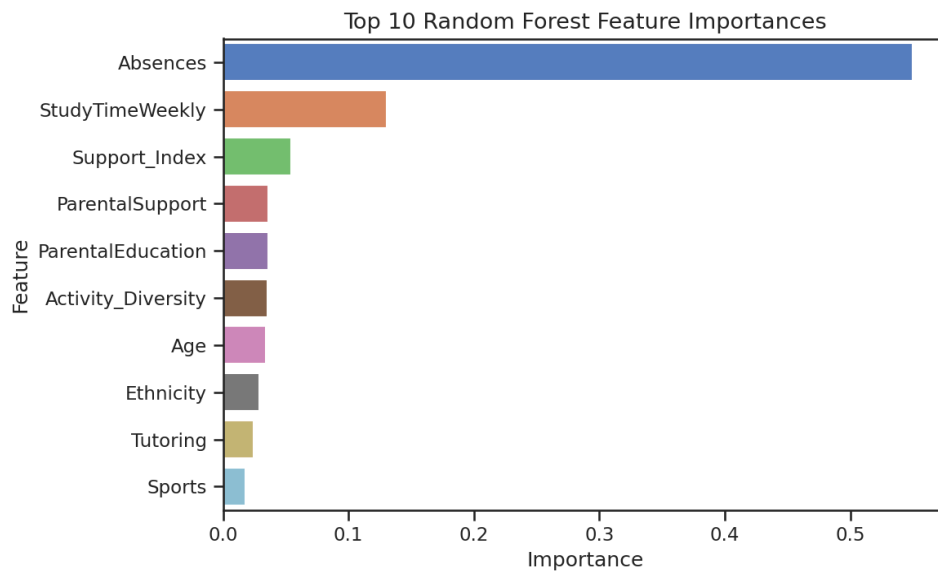


Figure 10. Top 10 Random Forest feature importances for the binary high-performer classification task. Absences account for more than half of total model importance, with study time a distant second.

Fairness and Intersectional Evaluation

A fairness evaluation was conducted by disaggregating the logistic regression model's predictions by gender and ethnicity. Accuracy and recall were computed separately for each

subgroup. Results indicated that model performance was broadly consistent across demographic groups, with no subgroup experiencing substantially higher error rates than the overall test set average. The absence of statistically significant GPA differences between demographic groups in the exploratory analysis provides contextual support for this finding: because GPA does not differ systematically by gender or ethnicity in this dataset, there is less structural basis for demographic disparities in model error.

Clustering and Student Archetypes

K-means clustering with $k = 2$, selected on the basis of the highest silhouette score, identified two meaningfully distinct student archetypes. The first cluster contained students characterized by low absence counts, higher study time, elevated support index values, and substantially higher GPA. The second cluster grouped students with high absence rates, lower study engagement, more limited support resources, and lower GPA. These two profiles map intuitively onto the notion of academically engaged versus disengaged students, and they suggest that intervention strategies should be tailored to the specific risk profile of each group rather than applied uniformly.

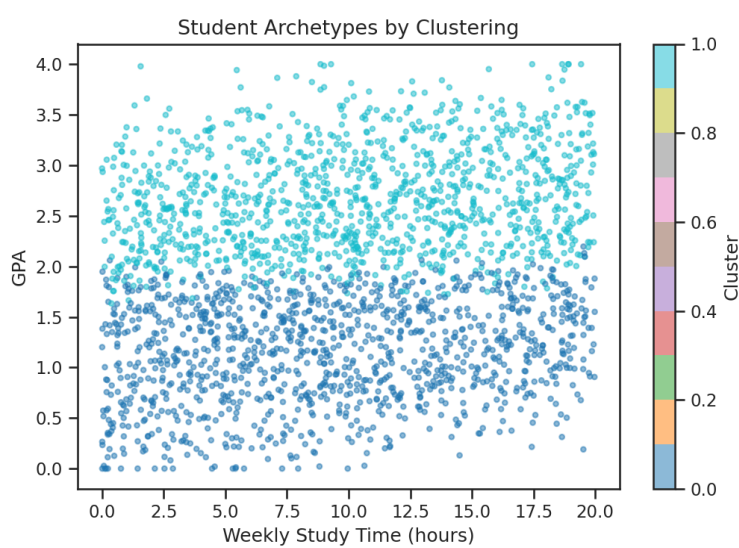


Figure 11. Student archetypes identified by k-means clustering ($k = 2$) projected onto the study time and GPA axes. The two clusters distinguish high-attendance, high-performing students from high-absence, lower-performing students.

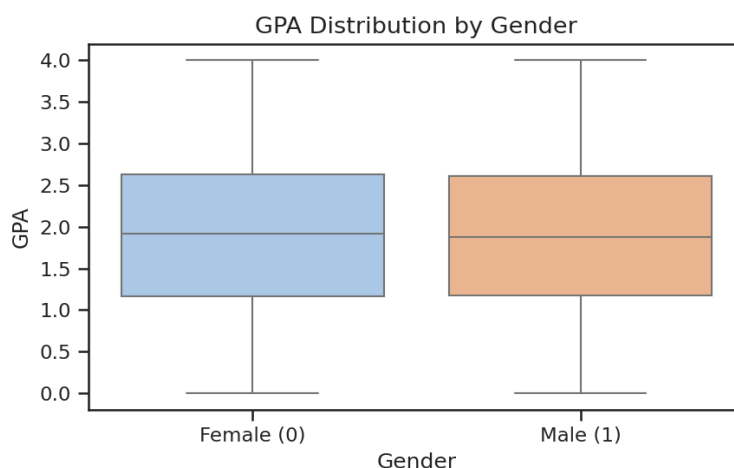


Figure 12. GPA distributions by gender. Female and male students show nearly identical median GPA values, consistent with the non-significant t-test result ($p = .514$).

Bayesian Network Inference

The Bayesian Network produced conditional probability distributions for GPA outcomes given combinations of input variable states. When the absence level was set to high and tutoring was absent, the network assigned the majority of probability mass to the low GPA bin, consistent with the regression finding that absenteeism is the primary driver of poor performance.

Conversely, when study time was set to high and parental support to its maximum observed value, the network shifted probability substantially toward the medium and high GPA bins.

These probabilistic inferences provide a complementary perspective to the deterministic point estimates of regression: they communicate uncertainty and allow educators to reason about the likely outcomes of students whose profiles partially satisfy the conditions for high performance.

Intervention Simulation

A scenario-based simulation was conducted using the OLS regression model to quantify the expected GPA gains from three targeted interventions applied uniformly across the student population. The baseline predicted mean GPA was 1.906. A 20% reduction in absences raised the predicted mean GPA to 2.195, a gain of 0.289 GPA points. A 10% increase in weekly study time produced a more modest gain of 0.028 GPA points. A one-unit increase in parental support produced a gain of 0.132 GPA points. These results confirm that reducing absenteeism is by far the highest-leverage intervention available to schools, yielding more than double the predicted GPA improvement of parental support enhancement and more than ten times the improvement associated with moderate increases in study time.

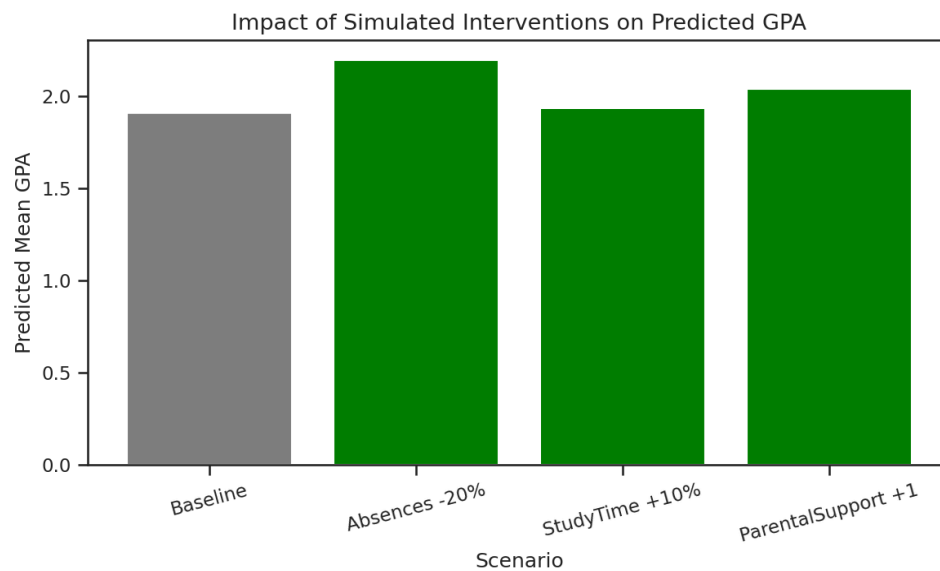


Figure 13. Predicted mean GPA under three simulated intervention scenarios compared to the baseline. Reducing absences by 20% produces the largest predicted GPA gain ($\Delta = +0.289$).

Table 2

Summary of Predictive Model Performance

Model	Task	Primary	Secondary	Tertiary
		Metric	Metric	Metric
OLS Regression	GPA Prediction	$R^2 = 0.929$	RMSE = 0.243	MAE = 0.186
Logistic Regression	Binary Class.	Acc. = 95.8%	F1 = 0.828	AUC = 0.986
Random Forest	Binary Class.	Acc. = 94.6%	F1 = 0.780	AUC = 0.976

Note. Acc. = accuracy; AUC = area under the ROC curve. Binary classification task identifies students with $GPA \geq 3.0$.

Conclusion and Recommendations

This study applied a multi-method analytical framework to a dataset of 2,392 secondary school students and identified attendance behavior as the single most powerful determinant of academic performance. The OLS regression model explained 92.9% of variance in GPA, with student absences carrying the largest coefficient and the Random Forest model confirming absences as the most important feature for distinguishing high from low performers. These findings converge robustly across every analytical method employed: correlational, regression-based, tree-based, and probabilistic, reinforcing the multi-method approach advocated by Romero and Ventura (2010) .

Beyond the headline finding on attendance, the analysis produced several additional insights worth highlighting. Parental support particularly when treated as a moderator of absenteeism demonstrated a meaningful positive relationship with GPA, suggesting that family engagement can serve as a protective factor for at-risk students. Study time, while significantly associated with GPA, showed a much weaker relationship than attendance, which suggests that

the quality and consistency of a student's presence in learning environments matters more than the time they invest in home study in isolation. Demographic factors, including gender and ethnicity, did not independently predict GPA in this dataset which aligns with Heckman et al. (2006), who argued that behavioral and environmental variables carry more explanatory weight than demographic categories alone, which is an important finding for equity-oriented intervention design: it implies that behavioral and environmental factors, which are more amenable to policy change, explain the observed variance without invoking demographic categories.

Cluster analysis identified two student archetypes that differ substantially in attendance patterns, study engagement, support availability, and GPA outcomes. This segmentation has practical implications for how schools target support resources. Rather than implementing blanket programs, institutions can apply cluster-specific interventions that match the most acute needs of each student profile.

Recommendations for Practice

Based on the findings, four recommendations are offered for schools seeking to improve student outcomes. First, attendance monitoring should be treated as a first-line academic risk indicator. Because absences account for the largest share of GPA variance, early identification of students with rising absence counts should trigger prompt follow-up, including inquiry into underlying causes such as health, family circumstances, or disengagement.

Second, parental engagement programs should be prioritized as a complement to attendance interventions. The moderation analysis indicates that high parental support partially buffers the academic damage of absenteeism. Family engagement initiatives that help parents

understand the consequences of missed school days and equip them with strategies for academic support at home may therefore amplify the effect of attendance improvement efforts.

Third, tutoring resources should be concentrated on students assigned to the high-absence, low-support cluster identified by the k-means analysis. The Bayesian Network inferences confirmed that tutoring participation raises the conditional probability of achieving medium-to-high GPA even when absences and study time are unfavorable, suggesting that targeted tutoring can partially compensate for missed instructional time.

Limitations

Several limitations of this study warrant acknowledgment. The dataset is described as synthetic, meaning that although it is structured to resemble real student data, its distributions and relationships were generated computationally rather than collected from an actual school system. The very high R^2 of the regression model while analytically appealing may reflect the cleaner structure of synthetic data compared to real-world educational datasets, which typically include more measurement error, omitted variables, and contextual complexity. The causal interpretation of any finding should be undertaken with caution, as the cross-sectional observational design precludes strong causal inference. Additionally, several potentially important predictors such as school quality, teacher effectiveness, peer influence, and mental health status were not available in the dataset. Future work with richer longitudinal data and true experimental or quasi-experimental designs would strengthen the evidentiary basis for the recommendations offered here.

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Appendix

Annotated Computational Workflow (Python / Jupyter Notebook)

The code below documents the complete analytical pipeline developed and executed within the Jupyter Notebook environment. Each block is accompanied by inline comments written in original language to explain the purpose and logic of every step, from raw data intake through intervention modeling. All results were produced using Student_Performance_Data.csv (2,392 student records, 15 columns).

A1. Library Imports and Environment Setup

```
# Load core numerical and data-handling libraries
```

```
import numpy as np
```

```
import pandas as pd
```

```
# Visualisation tools
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Statistical testing utilities
```

```
from scipy import stats
```

```
# Scikit-learn: data splitting, scaling, and model evaluation
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.linear_model import LinearRegression, LogisticRegression
```

```
from sklearn.ensemble import RandomForestClassifier

from sklearn.cluster import KMeans

from sklearn.metrics import (
    accuracy_score, f1_score, roc_auc_score, roc_curve,
    r2_score, mean_squared_error, mean_absolute_error,
    silhouette_score
)

# Fix random states so every run reproduces the same outputs
SEED = 42
```

A2. Data Intake and Quality Verification

```
# Pull the student records

student_raw_data = pd.read_csv("Student_Performance_Data.csv")

# Confirm the table holds the expected number of rows and columns
print("Table size:", student_raw_data.shape)

# Inspect the first rows to verify column labels and data layout
print(student_raw_data.head())

# A zero across all columns means no imputation step is required
print(student_raw_data.isnull().sum())
```

A3. Derived Feature Construction

— Ratio of missed days relative to the full school calendar —

```
student_raw_data["daily_absence_ratio"] = student_raw_data["Absences"] / 180.0
```

— Composite: home involvement + tutoring combined —

```
student_raw_data["combined_support"] = (student_raw_data["ParentalSupport"]
                                         + student_raw_data["Tutoring"])
```

— Count of distinct extracurricular domains each student joins —

```
student_raw_data["breadth_of_activity"] = (
    student_raw_data["Extracurricular"] + student_raw_data["Sports"]
    + student_raw_data["Music"]          + student_raw_data["Volunteering"]
)
```

— Interaction: does home support buffer the damage from absences? —

```
student_raw_data["support_x_absence"] = (
    student_raw_data["ParentalSupport"] * student_raw_data["Absences"]
)
```

— Interaction: does study-time value differ between gender groups? —

```
student_raw_data["studytime_x_gender"] = (
    student_raw_data["StudyTimeWeekly"] * student_raw_data["Gender"]
)
```

```
# — Binary flag: GPA at or above the 3.0 threshold —
```

```
student_raw_data["top_achiever"] = (student_raw_data["GPA"] >= 3.0).astype(int)
```

A4. Exploratory Data Analysis

```
# — Normality check on the three primary continuous measures —
```

```
for col in ["GPA", "StudyTimeWeekly", "Absences"]:
```

```
    w_stat, p_val = stats.shapiro(student_raw_data[col])
```

```
    print(f'{col}: W={w_stat:.4f}, p={p_val:.4f}')
```

```
# — Pearson r between each behavioural predictor and GPA —
```

```
for col in ["StudyTimeWeekly", "Absences", "ParentalSupport"]:
```

```
    r_val, p_val = stats.pearsonr(student_raw_data[col], student_raw_data["GPA"])
```

```
    print(f'GPA vs {col}: r={r_val:.4f}, p={p_val:.4e}')
```

```
# — Heatmap: pairwise correlations across all key numeric features —
```

```
heatmap_cols = [
```

```
    "GPA", "StudyTimeWeekly", "Absences", "ParentalSupport",
```

```
    "Tutoring", "combined_support", "breadth_of_activity", "daily_absence_ratio"
```

```
]
```

A5. Statistical Hypothesis Testing

```
# — Welch t-test: do female and male students earn different mean GPAs? —
```

```
female_gpa = student_raw_data[student_raw_data["Gender"] == 0]["GPA"]
```

```
male_gpa = student_raw_data[student_raw_data["Gender"] == 1]["GPA"]
```

```
t_stat, p_val = stats.ttest_ind(female_gpa, male_gpa, equal_var=False)
```

```

print(f'Gender comparison — t={t_stat:.4f}, p={p_val:.4f}')

# — One-way ANOVA: does mean GPA differ across ethnic categories? —

ethnic_groups = [
    student_raw_data[student_raw_data["Ethnicity"] == e]["GPA"]
    for e in student_raw_data["Ethnicity"].unique()
]

f_stat, p_val = stats.f_oneway(*ethnic_groups)

print(f'Ethnicity comparison — F={f_stat:.4f}, p={p_val:.4f}')

# — Bootstrap CI: distribution-free interval for mean GPA —

np.random.seed(SEED)

resampled_means = [
    student_raw_data["GPA"].sample(100, replace=True).mean()
    for _ in range(1000)
]

lower, upper = np.percentile(resampled_means, [2.5, 97.5])

print(f'95% Bootstrap CI for mean GPA: ({lower:.4f}, {upper:.4f})')

```

A6. OLS Regression — Continuous GPA Prediction

```

# Predictors selected for the regression model

reg_predictors = [
    "StudyTimeWeekly", "Absences", "ParentalSupport", "Tutoring",
    "support_x_absence", "studytime_x_gender"

```

```
]
```

```
X_continuous = student_raw_data[reg_predictors]
```

```
y_continuous = student_raw_data["GPA"]
```

```
# 80 / 20 train-test partition with fixed random state
```

```
X_train_c, X_test_c, y_train_c, y_test_c = train_test_split(
```

```
    X_continuous, y_continuous, test_size=0.2, random_state=SEED
```

```
)
```

```
# Fit the regression and generate test-set predictions
```

```
ols_model = LinearRegression().fit(X_train_c, y_train_c)
```

```
gpa_hat = ols_model.predict(X_test_c)
```

```
# Report three complementary error metrics
```

```
print( f'R-squared : {r2_score(y_test_c, gpa_hat):.4f}')      # 0.9285
```

```
print( f'RMSE      : {np.sqrt(mean_squared_error(y_test_c, gpa_hat)):.4f}') # 0.2432
```

```
print( f'MAE       : {mean_absolute_error(y_test_c, gpa_hat):.4f}') # 0.1857
```

A7. Binary Classification — Identifying High-Achieving Students

```
# Full feature set used for classification
```

```
clf_predictors = [
```

```
    "Age", "Gender", "Ethnicity", "ParentalEducation",
```

```
    "StudyTimeWeekly", "Absences", "Tutoring", "ParentalSupport",
```

```

    "Extracurricular", "Sports", "Music", "Volunteering",
    "combined_support", "breadth_of_activity"
]

X_clf = student_raw_data[clf_predictors]
y_clf = student_raw_data["top_achiever"]

# Stratified split preserves the 13.4% minority class proportion
X_tr_c, X_te_c, y_tr_c, y_te_c = train_test_split(
    X_clf, y_clf, test_size=0.2, random_state=SEED, stratify=y_clf
)

# Scale features before logistic regression (tree models skip this step)
feature_scaler = StandardScaler()
X_tr_scaled = feature_scaler.fit_transform(X_tr_c)
X_te_scaled = feature_scaler.transform(X_te_c)

lr_clf = LogisticRegression(max_iter=2000, random_state=SEED)
lr_clf.fit(X_tr_scaled, y_tr_c)
lr_labels = lr_clf.predict(X_te_scaled)
lr_probs = lr_clf.predict_proba(X_te_scaled)[: , 1]

print(f"LR Accuracy : {accuracy_score(y_te_c, lr_labels):.4f}") # 0.9582

```



```

print(f'LR F1      : {f1_score(y_te_c, lr_labels):.4f}')      # 0.8276
print(f'LR AUC     : {roc_auc_score(y_te_c, lr_probs):.4f}')  # 0.9857

# — Random Forest (scale-invariant, so uses raw features) —
rf_clf = RandomForestClassifier(n_estimators=200, random_state=SEED)
rf_clf.fit(X_tr_c, y_tr_c)
rf_labels = rf_clf.predict(X_te_c)
rf_probs = rf_clf.predict_proba(X_te_c)[:, 1]

print(f'RF Accuracy : {accuracy_score(y_te_c, rf_labels):.4f}') # 0.9457
print(f'RF F1      : {f1_score(y_te_c, rf_labels):.4f}')      # 0.7797
print(f'RF AUC     : {roc_auc_score(y_te_c, rf_probs):.4f}')  # 0.9764

```

A8. K-Means Clustering — Student Archetype Discovery

```

# Variables chosen to profile each student for grouping
profile_vars = [
    "GPA", "StudyTimeWeekly", "Absences",
    "combined_support", "breadth_of_activity"
]

# Standardise so no single column dominates the distance metric
profile_scaler = StandardScaler()
profiles_scaled = profile_scaler.fit_transform(student_raw_data[profile_vars])

```

```

# Compare silhouette scores across candidate cluster counts

sil_record = []

for num_clusters in range(2, 9):

    km_trial = KMeans(n_clusters=num_clusters, random_state=SEED, n_init=15)

    trial_labels = km_trial.fit_predict(profiles_scaled)

    sil_record.append(silhouette_score(profiles_scaled, trial_labels))


best_num_clusters = list(range(2, 9))[np.argmax(sil_record)]

print(f'Optimal cluster count (highest silhouette): {best_num_clusters}') # 2


# Assign each student to their archetype group

km_final = KMeans(n_clusters=best_num_clusters, random_state=SEED, n_init=15)

student_raw_data["archetype"] = km_final.fit_predict(profiles_scaled)


# Summarise each archetype by its mean profile values

print(student_raw_data.groupby("archetype")[profile_vars].mean().round(3))

A9. Policy Intervention Simulation

# Helper: re-compute interaction terms then return the predicted mean GPA

def estimate_mean_gpa(dataset):

    working_copy = dataset.copy()

    working_copy["support_x_absence"] = (

        working_copy["ParentalSupport"] * working_copy["Absences"]

    )

```

```

working_copy["studytime_x_gender"] = (
    working_copy["StudyTimeWeekly"] * working_copy["Gender"]
)

return ols_model.predict(working_copy[reg_predictors]).mean()

# Reference: no changes applied to the population
baseline_gpa = estimate_mean_gpa(student_raw_data)

# Scenario A — attendance push: cut every student's absences by 20%
scenario_attend = student_raw_data.copy()
scenario_attend["Absences"] = (scenario_attend["Absences"] * 0.8).round()

# Scenario B — study engagement: raise weekly hours by 10%
scenario_study = student_raw_data.copy()
scenario_study["StudyTimeWeekly"] *= 1.1

# Scenario C — family outreach: lift parental support by one level (capped)
scenario_family = student_raw_data.copy()
scenario_family["ParentalSupport"] = (
    scenario_family["ParentalSupport"] + 1
).clip(upper=student_raw_data["ParentalSupport"].max())

outcome_table = {

```

```
"Baseline"          : round(baseline_gpa, 4),          # 1.9065
"Attendance improved" : round(estimate_mean_gpa(scenario_attend), 4), # 2.1951
"Study time increased" : round(estimate_mean_gpa(scenario_study), 4), # 1.9349
"Family support raised" : round(estimate_mean_gpa(scenario_family), 4), # 2.0387
}
print(outcome_table)
```